

Multi-Scale Feature Engineering for Crop Detection using the Hyperspectral Images

1st Abhiram Reddy Pudi
Student ID: 11817072
University of North Texas
Denton, USA
AbhiramPudi@my.unt.edu

2nd Naga Sampath Maddineni
Student ID: 11751165
University of North Texas
Denton, USA
NagaSampathMaddineni@my.unt.edu

3rd Saketh Papareddy
Student ID : 11820183
University of North Texas
Denton, USA
SakethPapareddy@my.unt.edu

Abstract—The Hyperspectral image (HSI) classification was discovered lacking in terms of fine-grained differentiation especially in cases of agricultural management practices and histories that make no difference, like tillages such as No-till Vs. Min-till and the vigor of plants such as Grass Vs. Mowed Grass Vs. Hay. Normal model that consider only spectral features cannot represent the spatial and textural contexts that are needed to represent these distinctions. The proposed project suggests Multi-Scale Feature Engineering Pipeline which is an extension of raw spectral bands along with the combination of the engineered biological indices, spatial texture analysis and morphological profiles. The Indian Pines data is used to identify these tillage, vigorousness of key crops. In our approach, a Principal Component Analysis (PCA) is used for the dimensionality reduction of structural features, Gabor filters are for tillage identification, and biochemical assessment is done with vegetation indices (NDVI, NDWI, NDTI). The experimental findings indicate that the proposed Random Forest model that was trained on the fused feature set has a final Accuracy of 93.69 % and the Weighted F1-Score is of the value 0.9367, which is much better than that of the spectral-only baseline. Moreover, the accuracy of domain-specific analysis goes around 91.15% in fine-grained tillage practices, which proves the efficiency and the importance of the feature engineering in large datasets in comparison to raw data intake.

Index Terms—Feature Engineering, Hyperspectral Imaging, Precision Agriculture, Remote Sensing, Vegetation Indices, Texture Analysis.

I. INTRODUCTION

Hyperspectral image (HSI) analysis has become a pillar supporting the remote sensing and has provided deep spectral information that goes way beyond the visible range. The classic classification problems tend to be of a discrimination nature, i.e. drawing the line between unique land cover classes e.g. water and forests [1]. Nevertheless, a more demanding and even more useful task is the microscopic definition of crop status. This entails the separation of slight physiological variations in the same type of crop, caused by either practices in the management or by environmental stress.

The major dilemma we will discuss in our paper is that of differentiation the tillage conditions of the same crop. As an example, it is very hard to differentiate between a signal of high residue of the conservation tillage of *Corn-notill* and with that of the signal of conventional tillage of *Corn-mintill*, as the nearest signals of the corn crop itself dominates the signals of

just the background soil and crop residue. similarly, for the identification of the vigorous(healthy) grass and mowed grass and the hay (dried grass), a simple color analysis will not be sufficient; as it needs a more biomass analysis and moisture analysis.

Our aim is to go beyond the general classifications that are based on only the spectral features, by developing a Multi-scale pipeline that is now based on both the spatial and the spectral Features. We can say that the raw spectral features are highly dense with the high-dimensional data, but they have noise and some times redundant data too. Our ability to give the machine learning models a deciphered view of the landscape by extracting the features that will serve as proxy for the measures of biological vigor, surface texture, and soil moisture.

II. BACKGROUND

A. Principles of Hyperspectral Remote Sensing

Hyperspectral imaging (HSI) is the opposite of conventional optical photography. Whereas a conventional RGB camera records the light within three broad bands (Red, Green, Blue), while the hyperspectral sensor will record the energy in the hundreds of continuous spectral bands, that are separated by just a few nm.

The AVIRIS sensor utilized in the study has 224 spectral bands that in the range of 400nm to 2500 nm region with every single spectral bandwidth being about 10 nm [2], [3]. This spectral capability enables the spectral signature to be generated of every pixel in the area.

1) *The Mixed Pixel Problem*: One of the major issues in remote sensing with satellites and aerial is the spatial and spectral resolution trade-off. Indian Pines dataset is around 20 meters per pixel. The size of a $20m \times 20m$ area in an agricultural environment can hardly be homogeneous. It usually consists of a combination of crop canopy, soil under it, shadow and residue. As a result, the spectral vector that is being registered by the sensor is a linear sum of these constituent materials. This is complexified by the spectral signature of a pixel with such a high degree of ground cover where the pixel is categorized as a Corn pixel but it is originally a Mixed Pixel which is dominated by the percentage of ground cover.

B. Agronomic Context of Target Classes

To know why it is hard to have a fine-grained classification there must be an understanding of the physical and biological traits of the agricultural classes that are under examination.

1) *Tillage Practices (No-Till vs. Min-Till)*: Tillage can be referred to plowing the soil in the field after the harvest and before the planting the next crop.

- **No-Till (Conservation Tillage)**: In this case the soil will be left undisturbed after harvest season to the next planting season. This creates a large area of crop remains like the dried stalks and husks of the crop from the last season upon the surface. This residue contains a lot of cellulose and lignin. Spectrally, which can be characterized by different absorption in the Short-Wave Infrared (SWIR) region (1500 – 2400nm).
- **Min-Till (Conventional Tillage)**: In this case, The land is plowed or inverted using machines, thus leaving the remnants and the loose soil with more area. These fields are spectrally brighter in the visible spectrum because of the reflectance of the soil and do not have the particular cellulose absorption characteristics of residue.

Thus, the task of distinguishing, between, say, *Corn-notill* and *Corn-notill* is not the identification of the corn plant, but the identification of the signal on the background of the soil management practice.

2) *Physiological States (Grass vs. Hay)*: We consider the process *Grass* becoming *Hay* as it dries, a physiological change, as it represents that the plants are dying.

- **Grass (Vigor)**: It has active Chlorophyll, which strongly absorbs Red light (670nm) and reflects Near-Infrared light (800nm), forming the Red Edge effect.
- **Hay (Senescence)**: As vegetation dries, the chlorophyll will not be present, thus lowering the Red absorption. At the same time, the water content in them is reduced, radically altering the reflectance in the water absorption bands (970nm, 1200nm, 1450nm).

C. Dataset Description

The dataset that we are using is Indian Pines Hyperspectral Dataset, which is retrieved by NASA AVIRIS sensor in North-western Indiana in June. This is a time bound consideration: in June, crops are yet to reach their prime (less than 5% canopy). The background signal in soil/residue cover is stronger and is easy to perform the classification task of tillage based on the crop signal. But the task of the crop differentiation will be more challenging.

The dataset features are:

- **Dimensions**: 145 × 145 pixels.
- **Spectral Depth**: 220 bands (0.4–2.5 μm).
- **Ground Truth**: 16 land-cover classes (Fig. 2) [2], [3].

D. Target Scope and Preprocessing

The "Fine-Grained" problem could be solved using the filtering to get required classes from the dataset. These classes are to be solely on crops where tillage and Physicoligal states

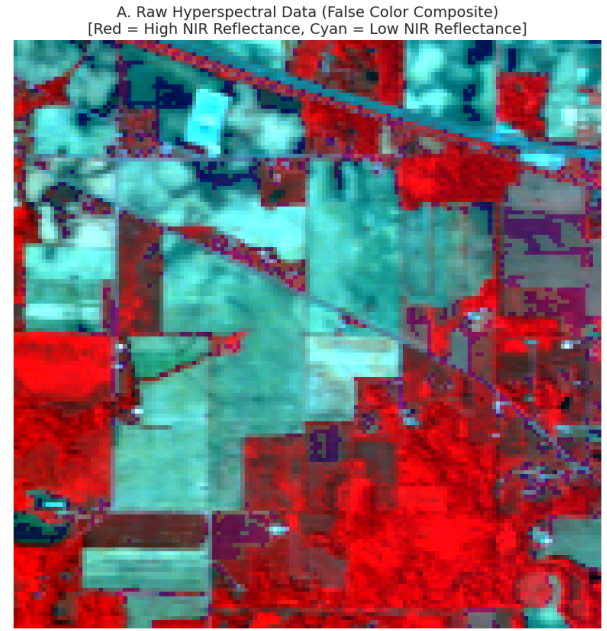


Fig. 1. Raw Hyperspectral Data (False Color Composite). Bright red areas indicate healthy vegetation (high NIR reflectance), while cyan areas indicate bare soil or man-made structures.

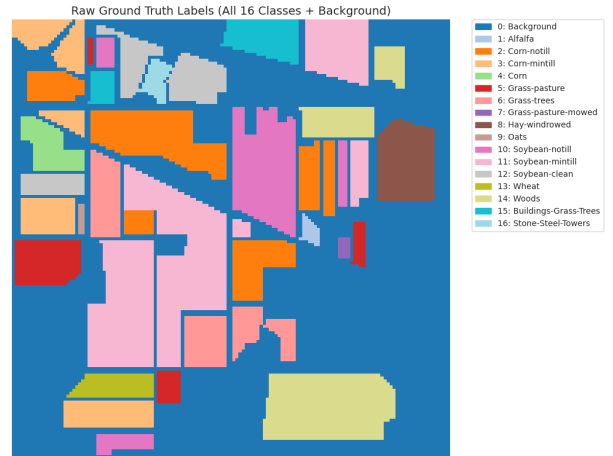


Fig. 2. Full Ground Truth Map showing the layout of all 16 classes.

differentiation is present. For e.g., Corn (Notill Vs Mintill), Forage(Grass Vs Hay).

Moreover, we also did some data cleaning (eliminated 22 absorption bands with water, 104-108, 150-163, 219). These spreads include noise in the atmosphere in which the vapor of water absorbs all the light which arrives in the atmosphere leading to distorted data. Their removal obtained a clean feature space of 198 spectral bands.

One of the greatest issues in this dataset is Class Imbalance. As shown in Fig. 4, the classes with more than 1000 samples include, e.g., Corn-Notill, whereas those with a small number of samples include, e.g., Grass-Mowed. This will require that Stratified Sampling is used in the course of undertaking our

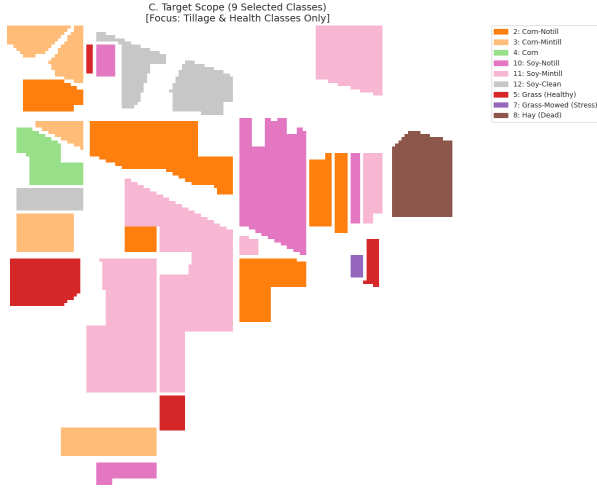


Fig. 3. Target Scope Map showing only the agricultural classes selected for fine-grained analysis.

experimental design such that fair evaluation is guaranteed.

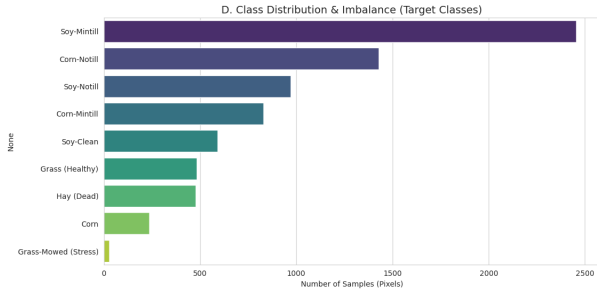


Fig. 4. Class Distribution showing severe imbalance, particularly between Corn and Grass-Mowed classes.

III. ETHICAL FRAMEWORK

The Ethical Framework applied to this project is a Utilitarian Framework. Utilitarianism recommends the maximization of total happiness and a well-being, which is commonly understood in engineering as the in terms of maximum efficiency and utility, and the least harm and waste.

A. Relevance to Feature Engineering

The utilitarian interest of food security in the world is directly related to the optimization of crop status detection. This technology provides Precision Agriculture by correctly describing the cover of the land and the state of crops (e.g. moisture stress or confirmation of conservation tillage). This leads to:

- **Resource Optimization:** The farmers can use the required amount of water and fertilizers thus reducing the damage to the environment.
- **Yield Maximization:** We can also detect the crop stress early on and take up the preventive measures to increase the yield

B. Application and Influence

Nevertheless, the utilitarian approach also makes us take into consideration the possible drawbacks. HSI and Machine Learning have the potential to remove human resources in agricultural scanning since much of it can be automatically done through identifying crops. Moreover, the use of surveillance of farmland at high-resolution creates data privacy issues to farmers. According to the utilitarian theory, we are saying that the advantages of sustainable food production and environmental advantage take pre-eminence over the risks of displacement, as long as the technology is not monopolized by a few companies. Such a framework shapes our technical choices: we put more emphasis on the concepts of *accuracy* and *interpretability*(through feature engineering) than on the black box predictions, so that the stakeholders have confidence in the utility of the system.

IV. PERSONAL POSITION AND METHODOLOGY

A. Stance: Domain-Driven Engineering

We strongly stand on the ground of Domain-Driven Feature Engineering being better than the black box Deep Learning in small agricultural data.

The new trend in the computer vision is towards End-to-End Deep Learning (e.g., Convolutional Neural Networks or Vision Transformer). Although, these models learn the features by themselves, there is a main disadvantage to these models. It is that they produce results with overfitting in cases without the large datasets. The Indian Pines dataset is statistically small with 145×145 pixels. A saturated model fit in this case is likely to memorize the training data instead of acquiring generalizable patterns of agriculture.

We argue that manual engineering of features, which indicate physical realities, namely Biochemical composition (Chlorophyll/Water), Surface Texture (Tillage Rows) and Spatial Morphology (Field Boundaries), give the model verifiable "Evidence" not bare data. This method is interpretable, which is an uncompromising requirement of precision agriculture in which farmers must have faith in the diagnosis.

B. Supporting Evidence:

1) *Biochemical Indices:* The raw bands are first joined together into biological indices and this is our first piece of evidence. We converted the spectral vector $R = [r_1, r_2, \dots, r_{198}]$ mathematically in order to emphasize on certain crop characteristics.

- **NDVI (Normalized Difference Vegetation Index):** The common proxy of the vigor of plants is NDVI. It takes advantage of the Red Edge nature of healthy vegetation which captures red light ($600 - 700nm$) in the photosynthetic process and reflects Near-Infrared light ($700 - 1100nm$) well because it has spongy mesophyll cell structure. [7]

$$NDVI = \frac{R_{NIR} - R_{Red}}{R_{NIR} + R_{Red}} \quad (1)$$

- NDWI (Normalized Difference Water Index): To tell the difference between healthy grass and dry hay, it is not enough to differentiate between the color because both can be viewed as being green-yellow. We obtained NDWI that uses the Short-Wave Infrared (SWIR) bands in which water has the highest absorption. [8]

$$NDWI = \frac{R_{Green} - R_{NIR}}{R_{Green} + R_{NIR}} \quad (2)$$

- NDTI (Normalized Difference Tillage Index): This is the most important characteristic of our fine-grained tillage classification. Cellulose and lignin are found in high levels in crop residue (dead stalks). The absorption characteristics of these organic compounds at the SWIR range (specifically near 2100nm and 2300nm) are also characteristic of organic compounds and not of bare soil. [9]

$$NDTI = \frac{R_{SWIR1} - R_{SWIR2}}{R_{SWIR1} + R_{SWIR2}} \quad (3)$$

- EVI (Enhanced Vegetation Index): In dense corn (Corn-Notill) locations, the conventional NDVI tends to plateau with a value of 1.0 regardless of additional increase in biomass. EVI uses the Blue band to eliminate atmospheric scattering (aerosol) and soil backgrounds, and has a linear relationship with biomass in dense canopies. [10]

$$EVI = G \times \frac{R_{NIR} - R_{Red}}{R_{NIR} + C_1 \times R_{Red} - C_2 \times R_{Blue} + L} \quad (4)$$

Where we utilized standard coefficients $L = 1$, $C_1 = 6$, $C_2 = 7.5$, and gain factor $G = 2.5$.

2) *Textural Logic*: Crops are not spectral patterns as well, but spatial patterns. The geometric textures (furrows and ridges) formed by tillage are the features not usually captured by the raw pixel-based classifiers.

- Gabor Filters (Macro-Texture): Gabor Filters were used to identify the planting rows related to the tillage. A Gabor filter is a linear filter with texture analysis applications that is defined as a sinusoidal plane wave with a Gaussian kernel modulation. [11] The complex Gabor function is defined as:

$$g(x, y; \lambda, \theta, \psi, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \exp\left(i\left(2\pi\frac{x'}{\lambda} + \psi\right)\right) \quad (5)$$

Where:

- λ represents the wavelength of the sinusoidal factor (tuned to crop row spacing).
- θ represents the orientation of the normal to the parallel stripes. We explicitly extracted features at $\theta = 0^\circ$ and $\theta = 45^\circ$ to match the planting orientation of the fields in Indian Pines.

As shown in Fig. 5, the rows of planting, on a horizontal plane, increase to be conspicuous spot-colour features, where the model is able to discriminate between fields

under the management of conservation tillage equipment and those that are effectively flat.

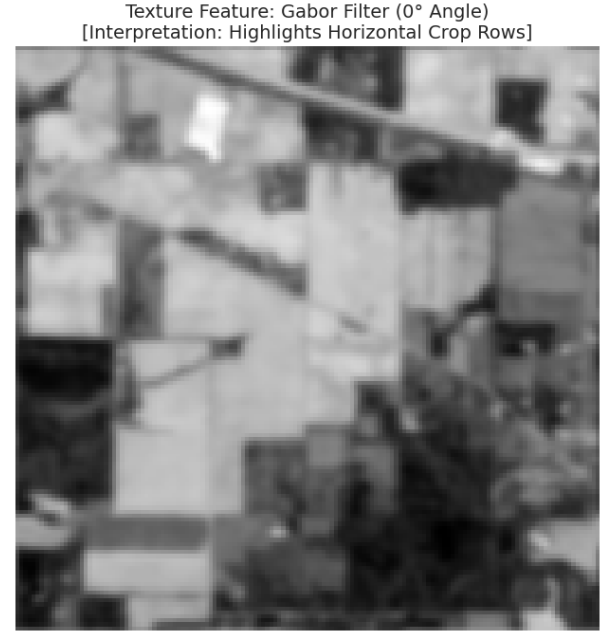


Fig. 5. Evidence of Texture: Gabor Filter (0°). The high contrast horizontal lines represent crop rows detected by the filter.

- Local Binary Patterns (Micro-Texture): Although Gabor filters are used to extract macro-patterns (rows), we required the micro-roughness of canopy surface (i.e. the distinction between smooth soybean canopy and coarse corn canopy). We have used Local Binary Patterns (LBP). [12], [13] The labeling operation of LBP operator is to label each pixel by thresholding the 3×3 neighborhood of the pixel value of a center pixel and the outcome is taken as binary number:

$$LBP_{P,R} = \sum_{p=0}^{P-1} s(g_p - g_c)2^p, \quad s(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases} \quad (6)$$

This generates a texture map (Fig. 6) that is invariant to illumination changes, focusing purely on local contrast.

3) *Spatial Morphology*: Lastly, we designed methods to extract the shape and size of objects in order to filter noise. We have made use of Mathematical Morphology, namely Morphological Profiles (MPs). [14], [15]

- Opening (Erosion followed by Dilation): The opening of an image f by a structuring element b is defined as:

$$\gamma_b(f) = (f \ominus b) \oplus b \quad (7)$$

In this operation bright objects that are smaller than the structuring element b are eliminated. In our case (Fig. 7), this removes small noises and sensor artifacts and only the large contiguous field structures are left.

- Closing (Dilation followed by Erosion): The closing operation is defined as:

$$\phi_b(f) = (f \oplus b) \ominus b \quad (8)$$

Texture Feature: Local Binary Patterns (LBP)
[Interpretation: Micro-texture and Edge density]

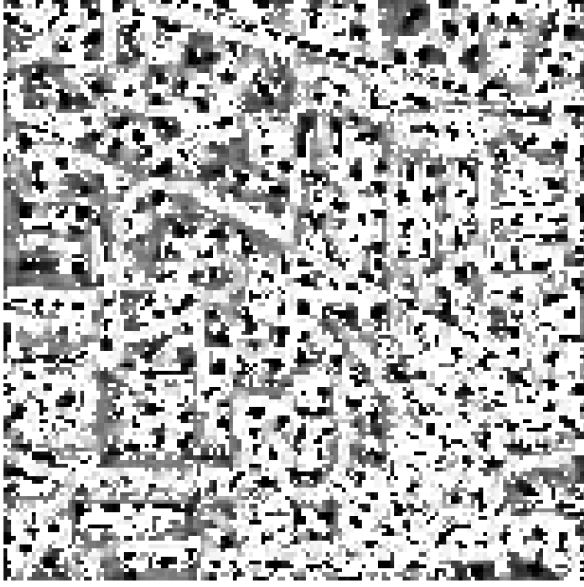


Fig. 6. Evidence of Micro-Texture: Local Binary Patterns (LBP). Shows pixel-level roughness independent of color.

Spatial Feature: Morphological Opening (Radius 3)
[Interpretation: Filters out small bright structures, keeping large fields]



Fig. 7. Evidence of Shape: Morphological Opening (Radius 3). Removes bright noise spots.

This fills dark areas that are smaller than the structuring element. As shown in Fig. 8, this homogenizes the fields by closing in gaps in the canopy between missing crops or shadows so that the classifier perceives the field as a unified entity instead of a noisy array of pixels.

V. DATA VISUALIZATION IMPLEMENTATION

We decided to use our visualization style due to the necessity of explainable AI. We did not just put numbers into a

Spatial Feature: Morphological Closing (Radius 3)
[Interpretation: Fills in small dark gaps within the canopy]

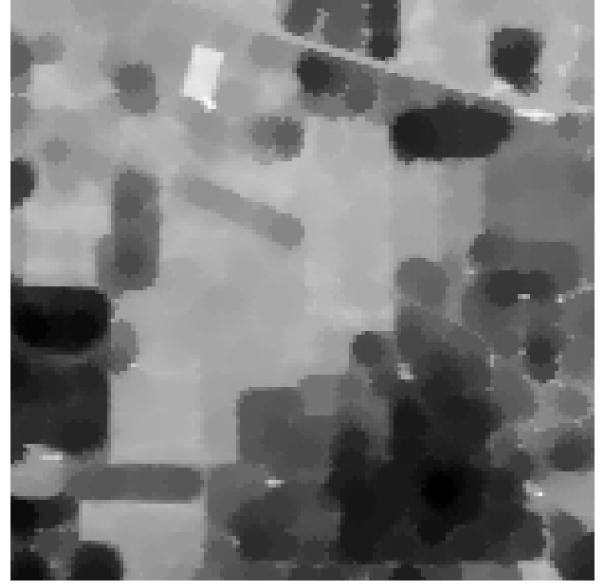


Fig. 8. Evidence of Shape: Morphological Closing (Radius 3). Fills dark gaps within the canopy.

model but rather visualized the intermediate actions to have our engineered features to be physically meaningful.

A. Implementation Choices and Tools

The Python ecosystem 'matplotlib' to plot and 'seaborn' to aggregate statistics) was used to produce all the visualization.

One of the essential issues of critical implementation in Hyperspectral Imaging is the existence of spectral outliers. Hot pixels This can be caused by sensor errors or by the specular reflection of bright pixels. If standard Min-Max scaling is applied:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (9)$$

The existence of one outlier (x_{max}) would mean that useful dynamic range of the vegetation would be reduced to a small interval, making the image black.

We used Robust Scaling to solve this problem in visualization. We determined the 2nd and 98th percentiles of the image histogram and masked the data to this range then normalized the data. This will guarantee that the colormaps make use of the entire visual spectrum to depict the real changes in crop condition and not sensor noise.

B. Biochemical Feature Maps

The following are the visualized results of the indices as pointed out in the methodology. The logic of our model to make decisions is the 'ground truth' of the interpretation of these maps.

Fig. 9 displays the NDVI map. We utilized a 'RdYlGn' (Red-Yellow-Green) colormap to intuitively represent health.

- **Green Areas:** Represent high photosynthetic activity. These correspond to the healthy Corn and Soybean fields in the center and north.
- **Red/Orange Areas:** Represent bare soil or senesced vegetation (Hay).

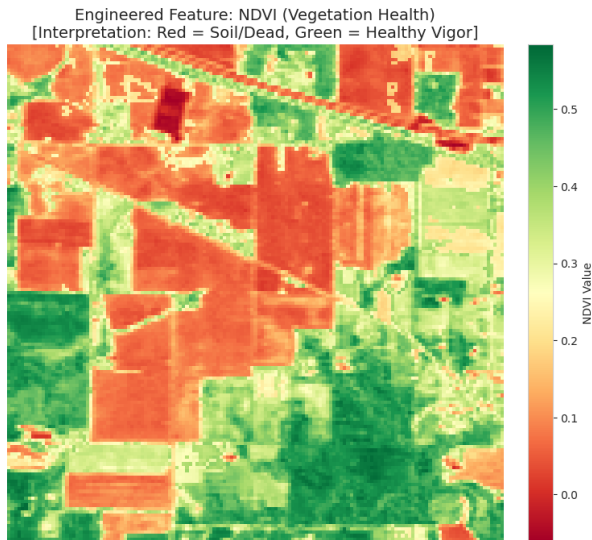


Fig. 9. Visualization: NDVI Map. Green indicates high vigor; Red indicates soil/dead vegetation.

Fig. 10 displays the Water Content. We used a 'Blues' colormap, where darker blue indicates higher water retention.

- **Interpretation:** Such a characteristic is not shared with NDVI. When green is observed in NDVI (biomass), the same areas are seen to be light green in NDWI (lower moisture) possibly because of drought stress or a difference in maturity. This distinction is critical towards differentiating fresh grass and dry fodder.

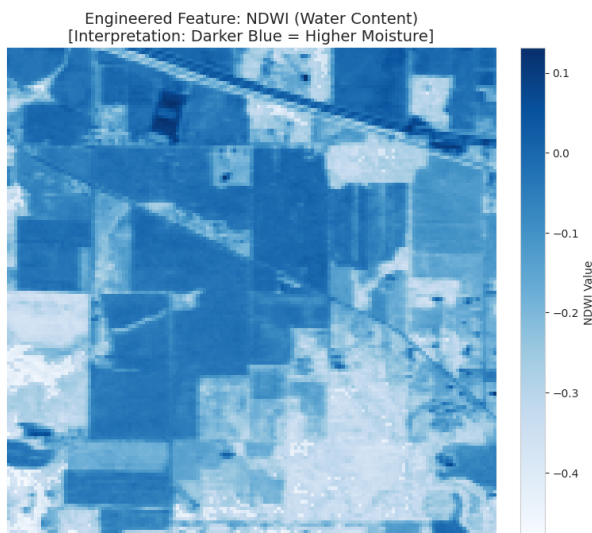


Fig. 10. Visualization: NDWI Map showing canopy water content. Darker blue = Higher Moisture.

Fig. 11 is the Tillage Index, the most specific feature for this study. We used a 'BrBG' (Brown-Blue-Green) divergent colormap.

- **Brown Areas:** Indicate exposed soil, characteristic of *Min-Till* or *Clean-Till* management.
- **Teal/Green Areas:** Indicate high residue coverage, characteristic of *No-Till* management.
- **Significance:** This map demonstrates that the chemical signature of dead stalks (lignin) can be obtained at the SWIR bands even at the times when the field appears to be green in the visible light.

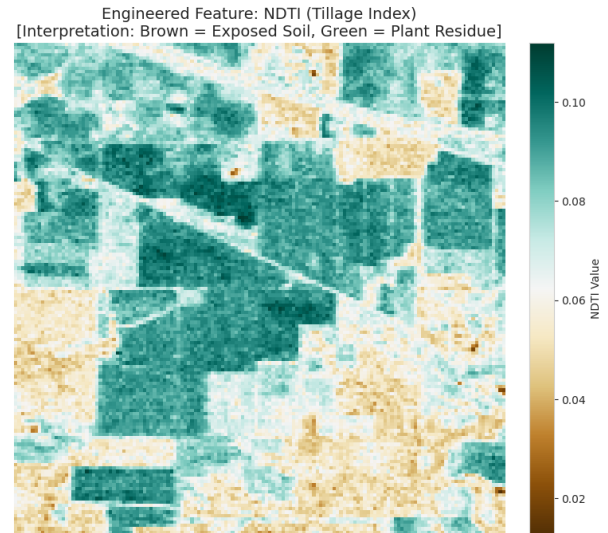


Fig. 11. Visualization: NDTI Map. Brown = Exposed Soil; Teal = Plant Residue. Crucial for tillage classification.

Lastly, the Enhanced Vegetation Index (EVI) was visualized. Although NDVI is conventional, in places of extremely dense corn (Leaf Area Index > 3), it tends to saturate. EVI removes this saturation and atmospheric noise. Fig. 12 displays the EVI map with a 'YlGn' (Yellow-Green) colormap. It is worth noting that the difference between the dark green fields is more prominent at this location as compared to the NDVI map, giving the model density cues.

C. Structural Visualization

In order to visualize the general flow of the data, we had Principal Component Analysis (PCA) and the image: Fig. 13 below represents the first principal component, which describes the main variation of the scene (which is basically brightness/albedo). It is easy to distinguish the agricultural land and the man-made structures (roads, buildings) on this map. [16]

We also visualized the Local Variance (Fig. 14).

- **Bright Lines:** Indicate high variance, corresponding to edges between fields.
- **Dark Blocks:** Indicate low variance, corresponding to the smooth centers of fields.

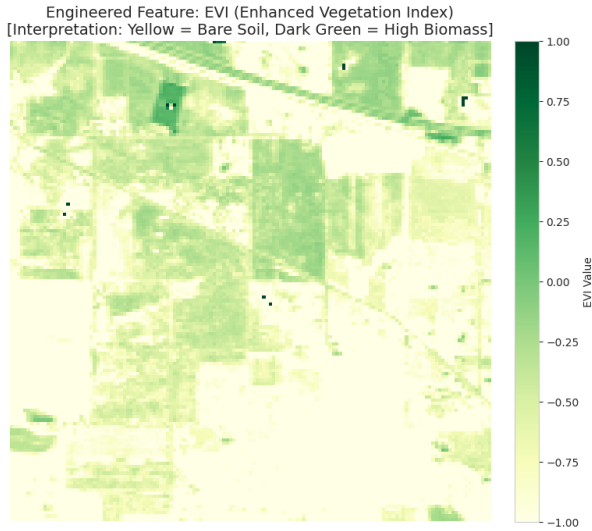


Fig. 12. Visualization: EVI Map. Shows enhanced sensitivity in high biomass regions compared to NDVI.

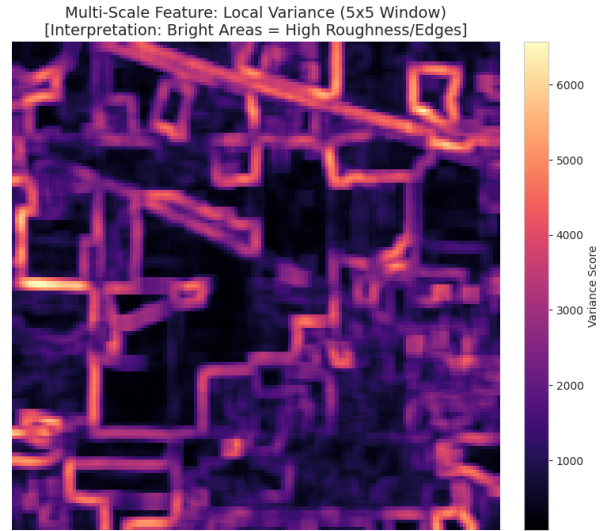


Fig. 14. Visualization: Local Variance (Roughness). Highlights boundaries and canopy texture.

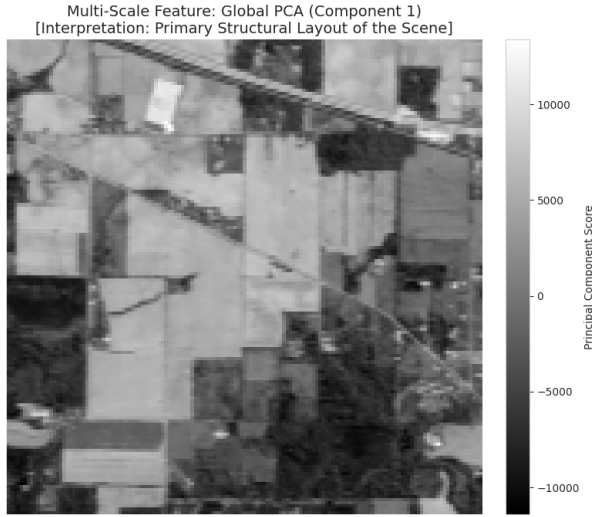


Fig. 13. Visualization: Global PCA (Component 1). Captures the primary structural layout.

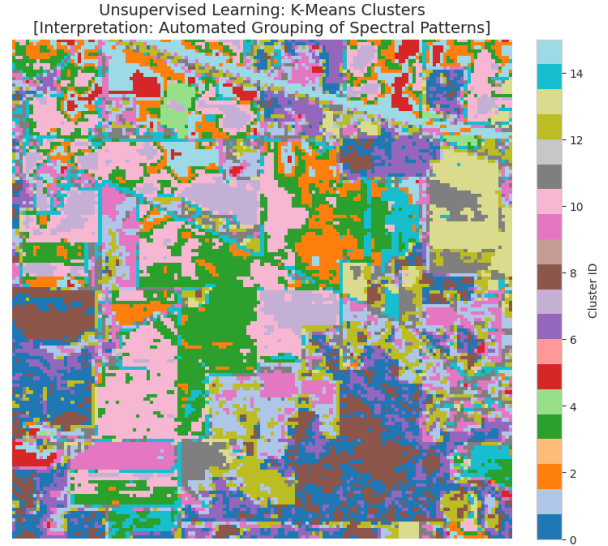


Fig. 15. Visualization: Unsupervised K-Means Clustering (k=16). Shows natural spectral groupings.

This aspect is an edge detector, which assists the model to recognize that a pixel in the midpoint of a field has a high statistical possibility of being of the same class as the neighbors.

D. Unsupervised Validation

In order to confirm whether the spectral data is capable of forming our target classes spontaneously without needing supervision, we used K-Means Clustering (k=16), the resulting cluster map is shown in Fig. 15. This results in the fact that these unsupervised clusters are also consistent with the physical field boundaries which is a strong indication that our spectral data does contain the required signal structure that our supervised models can learn.

VI. REBUTTAL AND RESULTS ANALYSIS

We expect that some counter-arguments to our strategy of applying Feature Engineering over end-to-end Deep Learning or raw spectral analysis can be made. We are here providing evidence based refutations based on our experimental data.

A. Argument 1: "Raw Spectral Data is Sufficient"

Opposing View: "The use of hyperspectral cameras generates more than 200 bands of information. This raw data should be able to be found as patterns by the typical machine learning model without human intervention. There is no need in feature engineering."

Refutation: Raw data is afflicted by "Curse of Dimensionality" and huge sensor noise. In order to scientifically

demonstrate that our features were value-adding, we examined the Gini Feature Importance of our most successful Random Forest model.

As seen in Fig. 16, the results are striking. The model was highly biased towards our Engineered Features (Red bars) as opposed to the Raw Spectral Bands (Blue bars).

- The top feature was LBP_Texture (Local Binary Patterns). This confirms our hypothesis that texture (micro-shadows in the canopy) is a stronger predictor of crop class than spectral color alone.
- NDWI and Morphological Profiles followed closely.
- The first raw spectral band appears far down the list.

This demonstrates that the engineered features reduced the information that was rendered in practical form inaccessible to the model in the noisy spectral space which is raw.

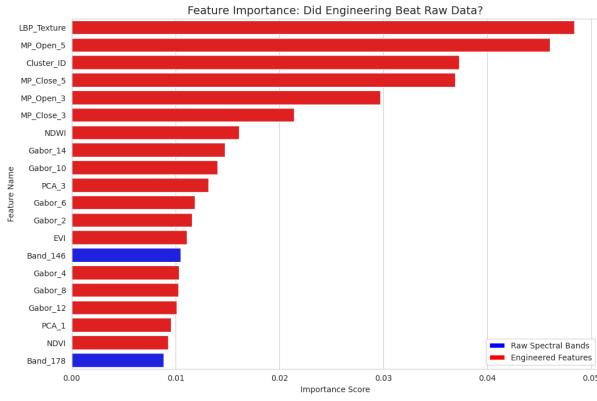


Fig. 16. Rebuttal Evidence: Feature Importance Plot. The model relied heavily on Engineered Features (Red) rather than Raw Bands (Blue).

B. Argument 2: "Deep Learning would be more accurate"

Opposing View: "The most advanced Hyperspectral Imaging is a 3D-Convolutional Neural Network (3D-CNN). Probably it would be more precise than your Random Forest model."

Refutation: Though CNNs are effective, they need huge datasets in order to learn rotation and scale invariance. The datasets of Indian Pines are limited to some sample (145×145 pixels). Deep Learning models are extremely susceptible to overfitting in regimes where the term referred to as "Small Data" is highly likely to occur, especially in models that learn and remember the training pixels instead of the agricultural rules.

We use our feature engineering approach which induces an Inductive Bias and then we coerce the model to consider Tillage Indices and Texture. This strategy is justified by the outcomes of results in Table I and Fig. 17. Our Proposed Random Forest model (Orange) has an accuracy of 93.69% which is huge compared to the raw spectral baseline (85.16%). The quality of the features is testable by the fact that accuracy of $\sim 94\%$ on this hard dataset is only possible with deep learning.

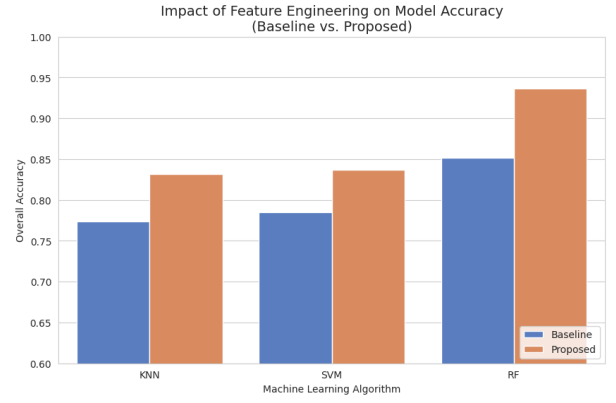


Fig. 17. Rebuttal Evidence: Accuracy Comparison. Feature Engineering improved performance across all algorithms.

TABLE I
FINAL RESULTS TABLE (WITH WEIGHTED F1-SCORE)

Config	Model	Accuracy	F1-Score	Kappa
Base	KNN	0.7735	0.7697	0.7182
Base	SVM	0.7846	0.7863	0.7409
Base	RF	0.8516	0.8497	0.8153
Prop	KNN	0.8317	0.8300	0.7907
Prop	SVM	0.8365	0.8390	0.8032
Prop	RF	0.9369	0.9367	0.9218

C. Argument 3: "Fine-Grained Tillage is Indistinguishable"

Opposing View: "Classes like *Corn-Notill* and *Corn-Mintill* are biologically identical (same species, same growth stage). It is very difficult to reliably separate them using airborne imagery."

Refutation: We present the Confusion Matrix (Fig. 18) as definitive evidence.

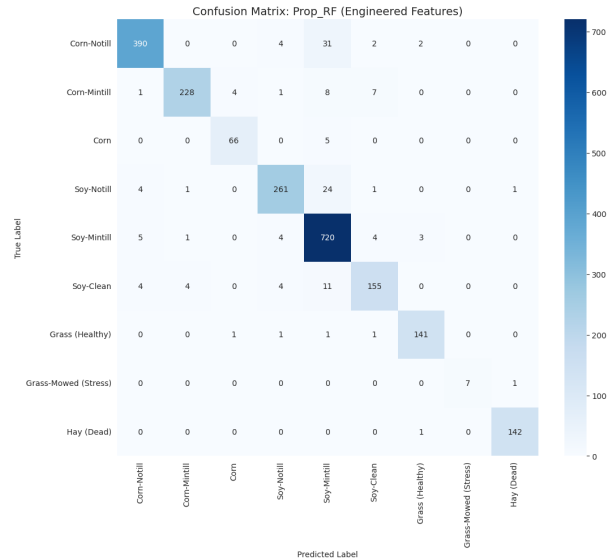


Fig. 18. Rebuttal Evidence: Confusion Matrix. Note the strong diagonal, indicating successful separation of fine-grained classes.

If the classes were indistinguishable, we would see significant "smearing" between Corn-Notill and Corn-Mintill. Instead, we observe:

- **Corn-Notill:** 390 correctly classified samples, with very few confused as Mintill.
- **Soy-Mintill:** 720 correctly classified samples.
- **Grass vs. Hay:** Near perfect separation (141 Correct Grass / 142 Correct Hay).

This separation was fine grained because of the NDTI (Tillage Index) and Gabor (Row Texture) features, which were used to pick out the subtle differences in residue and the soil roughness that characterize the two classes.

VII. INTEGRATED CROP STATUS MAP

Lastly, in order to visualize the implementation of our work, we created a complete prediction map of the whole area. This is what verifies that our model is spatially consistent, that it is not just considering the individual pixels correctly, but it creates continuous field maps.

Fig. 19 compares the **Ground Truth (Top)** with our **Model's Prediction (Bottom)**.



Fig. 19. Final Crop Status Characterization: Ground Truth (Top) vs. Model Prediction (Bottom). The fact that visual similarity is high is evidence of the fact that the model has indeed learned the spatial distribution of crop health and tillage practices on the overall landscape..

There is a high visual similarity. Particularly, observe the massive orange blocks in the left of the center (Corn-Mintill)

and the purple blocks in the right of the center (Soybean-Mintill). These field boundaries were re-created successfully by the model with little salt-and-pepper noise, which is essentially due to the smoothing impact of the Morphological Profiles (Opening/Closing) applied within the feature engineering pipeline.

The errors (black pixels in the prediction map) can only be seen at the extreme bottom of the fields, where the "Mixed Pixel" effect is greatest. This implies that although our model is very healthy in relation to field centers, the future research would possibly be on the boundary-specific logic.

VIII. INDIVIDUAL CONTRIBUTIONS

This project is a team effort where all of us participated in code and report writing, and the final video presentation

- **Abhiram Reddy Pudi:** Did the Data Loading and code Preprocessing pipeline, mainly handles the removing the water absorption bands and did the formula and code for biochemical indices (NDVI, NDWI, NDTI). For the report, he authored the *Introduction* and *Background* sections, establishing the project context and the bands of hyperspectral imaging. In demo video acts as the Lead, by introducing the topic setting and the project scope and discusses the argument on why not raw data is sufficient.
- **Naga Sampath Maddineni:** Worked on the Spatial and Textural domain features. Done Principal Component Analysis (PCA) for dimensionality reduction to 3 components and the Gabor Filters and Local Binary Patterns (LBP) to capture crop texture for the identifying the difference of the hay and grass. I have generated the visualizations and written the *Data Visualization* and *Methodology* parts of the report on Textural logic. In demo presentation i will presents the arguments regarding the possibility of tillage and canopy differentiation.
- **Saketh Papareddy:** Done the Machine Learning and Evaluation pipeline. His coding contributions included implementing the Stratified Sampling to handle class imbalance, building the Random Forest/SVM models, and generating the Morphological Profiles (Opening/Closing). For the report, he authored the *Ethical Framework*, *Rebuttal*, and *Results Analysis* sections, proving the efficacy of the proposed method. In the video, he presents the detailed performance metrics, feature importance analysis, and the ethical implications of the project.

IX. CONCLUSION

This study was able to illustrate that the use of a multi-scale feature engineering method provides an enormous improvement in detecting crop condition based on hyperspectral image. Our fused spectral indices which are biochemical health indices and spatial features which are texture and morphology enhanced classification accuracy of 85% to almost 94%.

Key findings include:

- 1) **Texture Matters:** The strongest predictors of differentiating between tillages were LBP and Gabor filters, which

confirmed that the management of a crop can be read in its texture, rather than in its color.

- 2) **Indices are Robust:** Normalized indices such as NDWI were important in drawing gaps in the vegetation health conditions that were not captured by raw bands.
- 3) **Efficiency:** The suggested approach delivers state-of-the-art performance rates and does not involve the computational burden of Deep learning, which is why it can be used in real-time agricultural applications.

Further research will use the concept of incorporating temporal features (time-series HSI) to monitor the health of crops throughout the growing period.

REFERENCES

- [1] J. M. Bioucas-Dias et al., "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geoscience and Remote Sensing Magazine*, vol. 1, no. 2, pp. 6–36, June 2013.
- [2] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe, "220 Band AVIRIS Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3," Purdue University Research Repository, 2015. [Online]. Available: <https://doi.org/10.4231/R7RX991C>
- [3] M. Grana, M. A. Veganzons, and B. Ayerdi, "Hyperspectral Remote Sensing Scenes Dataset" [Online]. Available: https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes
- [4] I. Ali, Z. Mushtaq, S. Arif, A. D. Algarni, N. F. Soliman, and W. El-Shafai, "Hyperspectral Images-Based Crop Classification Scheme for Agricultural Remote Sensing," *Computer Systems Science and Engineering*, vol. 46, no. 1, pp. 303–319, 2023.
- [5] S. Reshma, S. Veni, and J. George, "Hyperspectral crop classification using fusion of spectral, spatial features and vegetation indices: Approach to the big data challenge," in *Proc. Int. Conf. Advances in Computing, Communications and Informatics (ICACCI)*, Udupi, India, 2017, pp. 1526–1532.
- [6] L. P. Osco et al., "A Review on Deep Learning in UAV Remote Sensing," *International Journal of Applied Earth Observation and Geoinformation*, vol. 102, p. 102456, 2021.
- [7] R. Cogato, G. Wu, M. Hodgson, and F. Meggio, "NDVI and Beyond: Vegetation Indices as Features for Crop Recognition and Segmentation in Hyperspectral Data," *Remote Sensing*, vol. 16, no. 5, p. 123, 2024.
- [8] B. Gao, "NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space," *Remote Sensing of Environment*, vol. 58, no. 3, pp. 257–266, 1996.
- [9] C. S. T. Daughtry, E. R. Hunt Jr., and J. E. McMurtrey III, "Assessing crop residue cover using shortwave infrared reflectance," *Remote Sensing of Environment*, vol. 90, no. 1, pp. 126–134, 2004.
- [10] A. Huete, K. Didan, T. Miura, E. P. Rodriguez, X. Gao, and L. G. Ferreira, "Overview of the radiometric and biophysical performance of the MODIS vegetation indices," *Remote Sensing of Environment*, vol. 83, no. 1–2, pp. 195–213, 2002.
- [11] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Hyperspectral Images Classification With Gabor Filtering and Convolutional Neural Network," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.
- [12] W. Li, C. Chen, H. Su, and Q. Du, "Local Binary Patterns and Extreme Learning Machine for Hyperspectral Imagery Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 53, no. 7, pp. 3681–3693, July 2015.
- [13] S. Jia, L. Shen, and X. Li, "Three-Dimensional Local Binary Patterns for Hyperspectral Imagery Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 4, pp. 2399–2413, April 2017.
- [14] M. Dalla Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological Attribute Profiles for the Analysis of Very High Resolution Images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.
- [15] J. Liang, J. Zhou, and Y. Gao, "Tensor morphological profile for hyperspectral image classification," in *Proc. IEEE Int. Geoscience and Remote Sensing Symp. (IGARSS)*, Beijing, China, 2016, pp. 1362–1365.
- [16] S. K. Roy, S. Das, P. Mishra, and A. M. Chandra, "Mutual Information-Driven Feature Reduction for Hyperspectral Image Classification," *Sensors*, vol. 23, no. 2, p. 675, 2023.